

Firm- and Regime-Conditioned NLP Alpha

Overcoming three structural limits of current text-to-signal pipelines

Mingyang Liu • YL7919@ic.ac.uk • Independent research memorandum • June 2026 • Tags: **PRIOR-LIT** prior literature **OURS** our contribution **FUTURE** future

1. Research Question

Can text-derived equity signals be improved by conditioning language on firms and market regimes, and by replacing rank-only outputs with calibrated estimates of expected return and confidence?

Current NLP pipelines often assume that language has the same return implication across firms and market environments. This proposal relaxes these assumptions and focuses on producing more informative, robust, and production-ready alpha signals.

The scope is strictly **text** → **NLP** → **signal** → **delivery**; portfolio construction, risk management, optimisation, and execution remain with downstream teams.

2. Why Current Methods Are Insufficient

Across replications of supervised sentiment (SESTM), dictionary methods (LM), commercial feeds (RavenPack-style), and embedding-based return models, three assumptions recur — and each is often strained in empirical implementation.

- **Firm-invariant language.** A single word-to-return mapping is effectively applied market-wide. But “restructuring”, “dilution”, “activist”, or “cost reduction” are favourable for some firms and adverse for others; a global map books the average and mislabels the tails — exactly the idiosyncratic names where text matters most.
- **Time-invariant language.** Word polarity is treated as stable. In practice the *return relevance* of language shifts with the regime: balance-sheet and margin language is priced differently in a tightening or risk-off environment than in an expansion. Rolling re-estimation reacts late and can degrade performance around regime transitions.
- **Rank-only output.** The signal is typically a cross-sectional score used to sort securities into deciles. While useful for ranking, it does not provide calibrated estimates of expected return or confidence for downstream decision-making.

Why it matters: a senior NLP audience already knows the headline benchmark — an equal-weight long–short Sharpe on the order of 4.3 on daily news sentiment (Ke, Kelly & Xiu, 2019) **PRIOR-LIT** — is gross, high-turnover, and small-cap concentrated, and that reported LLM-based alpha can be materially overstated under contaminated or non-point-in-time evaluation **PRIOR-LIT**. The remaining opportunity lies in how textual information is conditioned, calibrated, and translated into signals.

3. Proposed Research

Four directions, each relaxing an assumption above or hardening the pipeline; models and derivations are in the Appendix.

1 Firm-conditioned sentiment, OURS

Condition sentiment on the firm at the *topic/event-cluster* level (clusters, not raw words, to keep estimation feasible): $s_{i,c} = s_c + e_i^\top W h_c$ with low-rank W , e_i a firm characteristic embedding and h_c a cluster embedding, shrunk toward the market map. This allows the same topic to have different return implications across firms.

Why it matters: recovers sign in names a market-wide map gets wrong; particularly relevant for small and idiosyncratic firms.

3 Ranking to calibration OURS

Calibrate scores into an expected-return estimate $\hat{\mu}_i$ with a confidence $\hat{\sigma}_i$, using heavy-tail-aware objectives (Huber / quantile / Student- t) with isotonic calibration rather than a Gaussian default. Output the pair $(\hat{\mu}_i, \hat{\sigma}_i)$, not a rank.

Why it matters: delivers a calibrated, confidence-aware signal that downstream teams can consume directly — the agenda stops at signal generation.

2 Regime-conditioned language OURS

Let sentiment vary with an *ex-ante* macro state z_t (volatility, rates/curve, cycle) through an additive cluster-level term, $s_{i,c,t} = s_c + e_i^\top W h_c + \beta_c^\top z_t$. Regimes are continuous and constructed only from observable information available at t , avoiding both look-ahead bias and latent-state estimation.

Why it matters: higher in-regime hit rates and more stable behaviour through regime transitions — the signal adapts when the return relevance of language changes.

4 Production alpha pipeline OURS

A point-in-time workflow: multi-source text → processing → features → signal → validation → delivery, with entity resolution, novelty scoring, near-duplicate news filtering, leakage screens, factor-neutrality checks, and signal-decay monitoring as first-class steps.

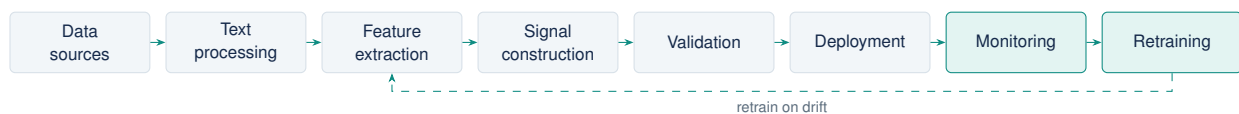
Why it matters: turns a research signal into a maintainable, monitorable production feed (§5).

4. Expected Contribution

- A **firm- and regime-conditioned sentiment estimator** that is interpretable and explicitly conditioned on observable firm and market characteristics, rather than relying solely on larger black-box models.
- A **calibrated, sized signal** $(\hat{\mu}_i, \hat{\sigma}_i)$ with a documented delivery contract, replacing rank-only output.
- A **reproducible, point-in-time research pipeline** with leakage and decay controls, plus replication baselines (SESTM, LM, commercial feed, embeddings) to attribute every gain.

5. Production Relevance

The signal is engineered for hand-off, not for our own backtest. The pipeline is modular and point-in-time end to end:



Inputs contribute at different horizons — **newswire** (Dow Jones / Bloomberg) and **RavenPack** for low-latency events; **earnings-call transcripts** (FactSet) and **SEC filings** for firm-specific language and formal disclosure; **alternative text** for breadth (full table in the Appendix). The hand-off is a versioned contract — *signal id*, *as-of timestamp*, *horizon*, $\hat{\mu}_i$, *confidence*, *coverage*. Crucially the loop does not end at delivery: **monitoring** for concept drift and scheduled or triggered **retraining** keep the signal calibrated as language and markets evolve.

6. Future Extensions

From current practice, kept only where they fit our mandate (detail and references in the Appendix):

- **Information-decay modelling** FUTURE — learn event-specific, regime-dependent alpha-decay curves (half-lives), so that each signal is accompanied by an estimated information horizon rather than a directional score alone; estimable with simple exponential/piecewise kernels first.
- **Blind-input validation** FUTURE — score entity-anonymised news (“Company A”) to test whether alpha is textual content or company/ticker memorisation; a diagnostic, not a signal.
- **LLM representations & point-in-time RAG** FUTURE — decoder-LLM embeddings as interpretable-layer features; retrieval over as-of-dated corpora with temporal-violation screens to keep LLM use backtestable.
- **Event-aware & agentic tooling** FUTURE — condition on event type, fuse modalities; multi-agent systems as gated research accelerators, not signal sources.